CUSTOMER SALES ANALYSIS

Week 5 — Advanced Data Manipulation with Pandas

Project Documentation & Analysis Report

1. Project Overview

This project applies advanced Pandas operations to sales and customer churn practice datasets. It investigates product, regional, monthly, and customer-level sales summaries; performs multiple-condition filtering; transforms text and date columns; demonstrates a dataset merge; and creates a pivot table.

2. Objectives

Analyze customer purchasing patterns, identify top customers and sales categories, summarize monthly performance, explore churn data, and produce an organized set of reproducible CSV outputs and charts.

3. Dataset Description

sales_data.csv contains 100 rows and seven columns (Date, Product, Quantity, Price, Customer_ID, Region, Total_Sales). customer_churn.csv contains 500 rows and nine columns (CustomerID, Tenure, MonthlyCharges, TotalCharges, Contract, PaymentMethod, PaperlessBilling, SeniorCitizen, Churn).

4. Data Preparation and Methods

Sales date values are parsed with pandas.to_datetime. String fields are trimmed and normalized with .str methods. Numeric columns are converted safely; incomplete and invalid records are filtered, and duplicate rows are removed. Date parts Year, Month, Day, and Month_Label are extracted.

5. Advanced Pandas Operations

Grouping uses sum, mean, and count aggregations by product, region, month, and customer. Filtering examples include an AND condition (North region and sales above/equal to median) and an OR condition (Laptop/Phone product or West region). A Region-by-Product pivot table uses pd.pivot_table with sum aggregation.

6. Dataset Merge and Important Key Note

The sales customer identifiers use the pattern CUST### while the churn dataset uses C#####. No exact common customer key is supplied. The project constructs a sequence crosswalk to demonstrate merge() mechanics; this linkage is artificial and is not a verified customer identity match. The churn profile and churn rate are reported independently from the churn dataset.

7. Key Findings

100 valid sales rows; total sales 12,365,048.00; average sales per row 123,650.48; Laptop is top product (3,889,210.00); North is top region (3,983,635.00); March 2024 is the strongest month (4,485,006.00). Churn dataset churn rate is 10.60%.

8. Business Interpretation

Laptop and North lead sales value in this practice dataset, and March has the strongest monthly value. A real business could investigate stock levels and regional campaigns, while customer retention insights should only be considered after a valid customer key and repeat purchase history are provided. These results are descriptive rather than causal.

9. Code Structure

main.py runs cleaning, group aggregations, filters, pivot generation, the documented demonstration merge, exports, charts, and summary printout. customer_analysis.ipynb offers a cell-by-cell learning path. data/ contains source CSVs; outputs/ contains CSV summaries; visualizations/ contains PNG charts; reports/ contains this report.

10. Setup and Installation

Install Python 3.10+ and project dependencies with: python -m pip install -r requirements.txt. Run the project from VS Code terminal using: python main.py. If needed, use the interpreter-specific launcher command (for example py -3.13 -m pip install -r requirements.txt, then py -3.13 main.py).

11. Technical Requirements Addressed

Pandas is used for manipulation throughout; three or more aggregation operations (sum, mean, count) are included; merge/join is demonstrated; pivot table summarization is implemented; and Matplotlib charts cover product, month, region, churn, and region-product pivot categories.

12. Deliverables

main.py, customer_analysis.ipynb, both source datasets, requirements.txt, README.md, CSV summary outputs, five professional PNG visualizations, and this PDF report.

13. Limitations

Sales dataset has 100 distinct customer IDs and only one sales row per customer, so repeat customer behavior and retention cannot be estimated from it. The merge crosswalk is artificial. Profit cannot be calculated because costs, returns, and discounts are not supplied.

14. Visual Evidence

Figure 1 — Sales by product

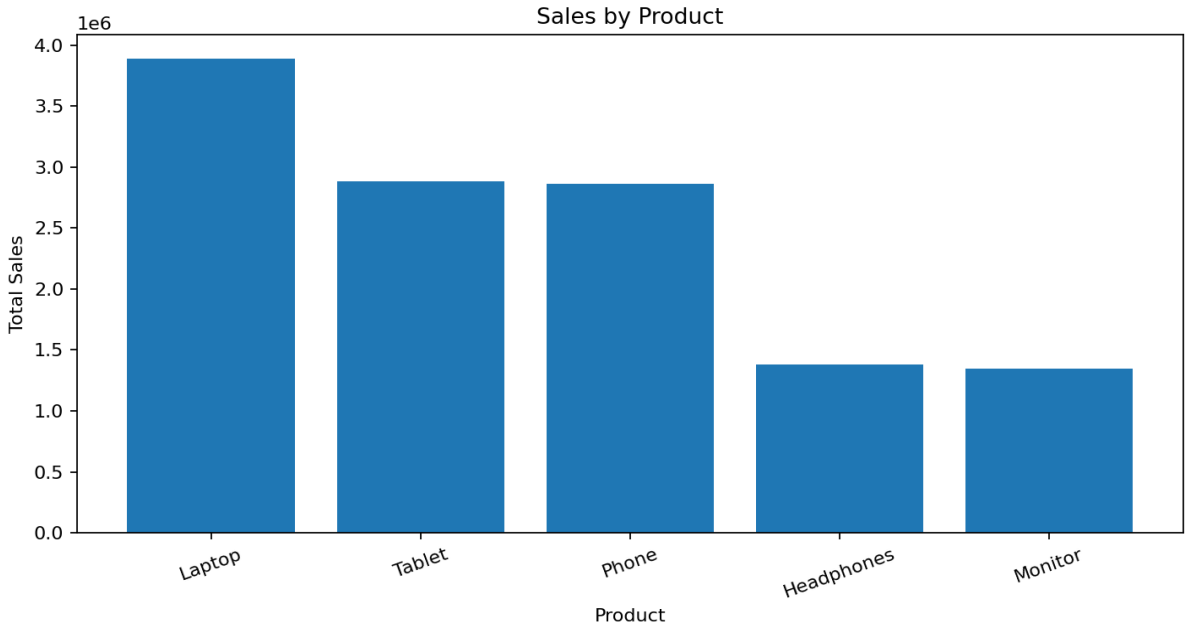


Figure 2 — Monthly sales trend

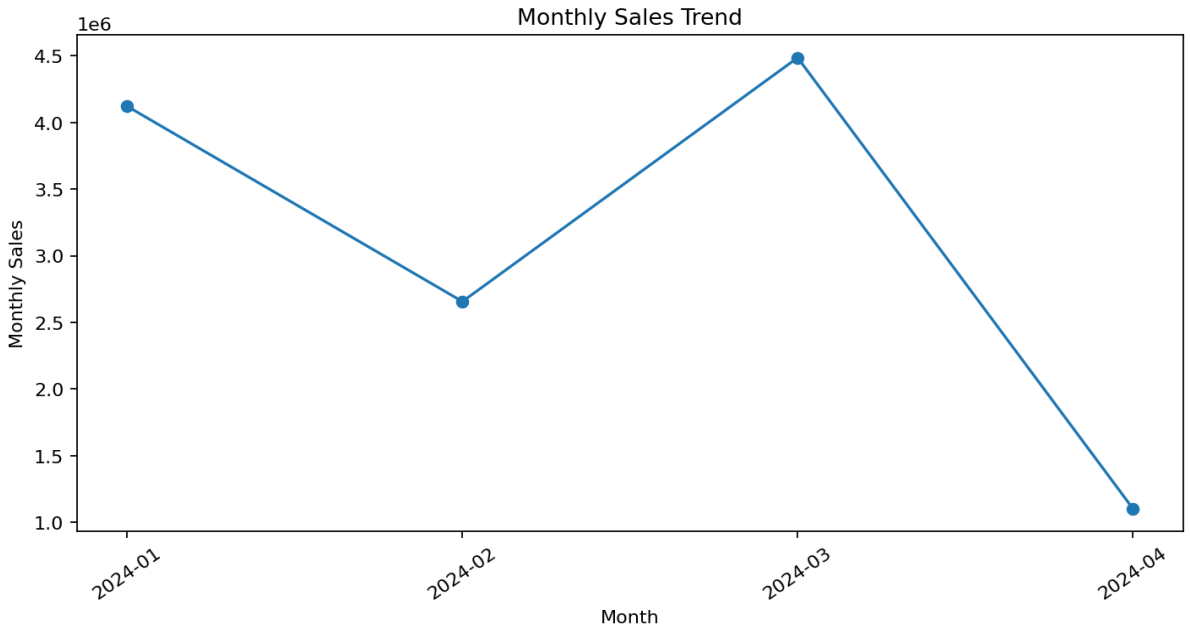


Figure 3 — Sales by region

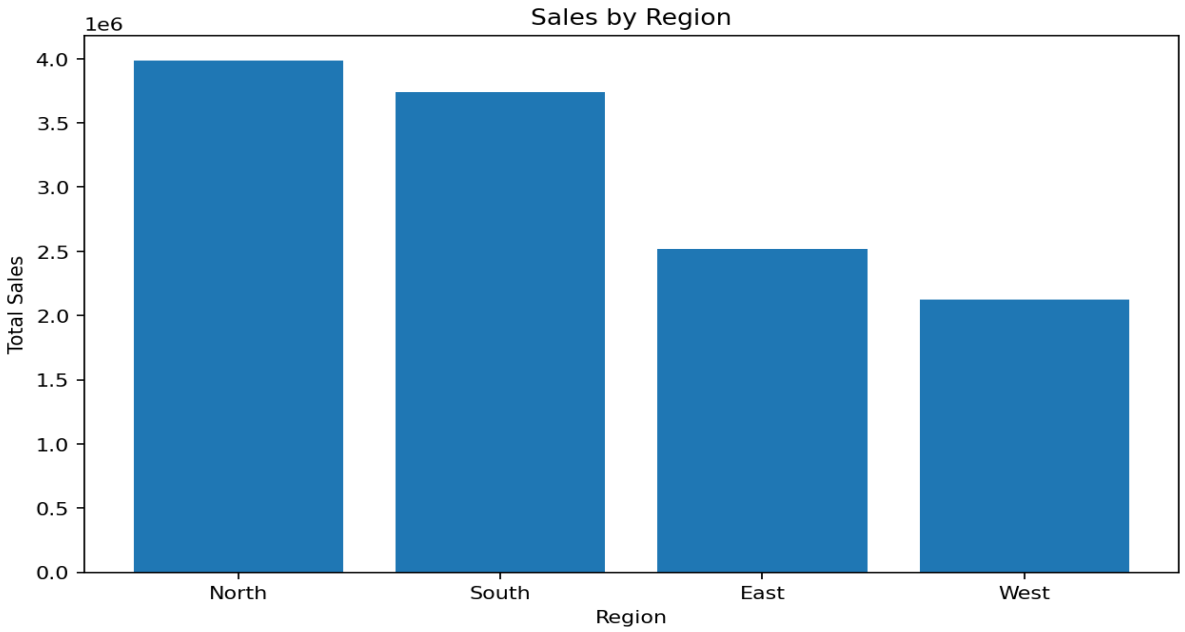


Figure 4 — Churn distribution

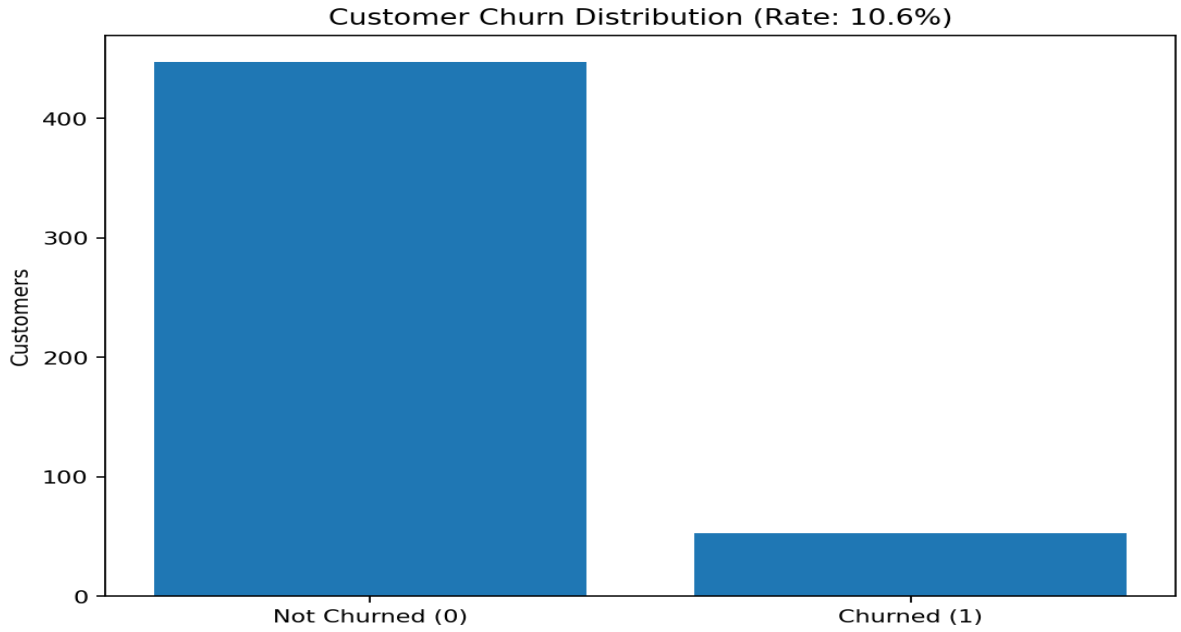
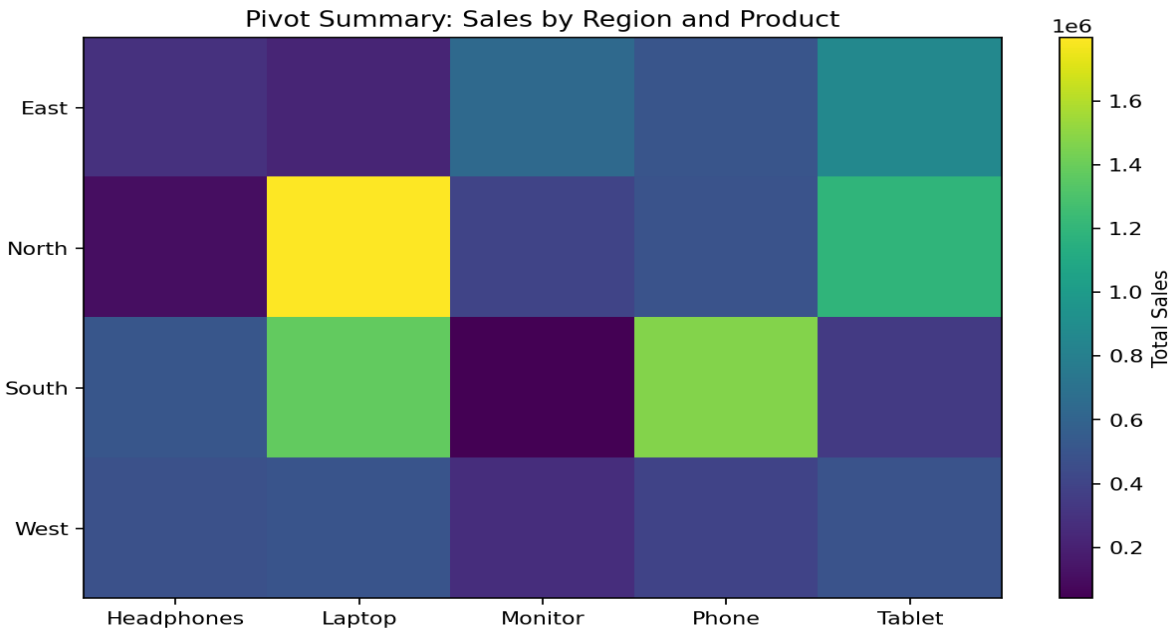


Figure 5 — Pivot table visualization


Pivot Summary: Sales by Region and Product
East
North
South
West
Headphones
Laptop
Monitor
Phone
Tablet
1e6
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
Total Sales